UNIT 2 · STANDARD 6.DS.6C

Describe Data by Mean Absolute Deviation

Lesson 2-9

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.

Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean Absolute Deviation | **Deviation** | **Absolute Value** | **Spread** | **Data distribution** | **Variability** | **Mean** | **Variation**

Level 3: try the blanks from memory first, then check the bank.

1. The average distance of each value from the mean is the _____.

2. How far a single value is from the mean is its _____.

3. The distance of a number from zero, never negative (zero is allowed), is its _____.

4. How far apart the data values are from each other is the _____.

5. The way data values are spread out is the _____.

6. How much the data values differ is the _____.

7. The total shared equally among all the values is the _____.

8. How much the values differ from one another is the _____.

Write About the Math

Both keepers allow the same average goals. Which should the coach pick and why?
Los dos porteros permiten el mismo promedio de goles. ¿A cuál debe escoger el entrenador y por qué?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Mean Absolute Deviation** *Desviación media absoluta*
- ☐ **Deviation** *Desviación*
- ☐ **Absolute Value** *Valor absoluto*
- ☐ **Spread** *Dispersión*
- ☐ **Data distribution** *Distribución de datos*

Model: *Absolute value makes each deviation a nonnegative distance, including zero, so signed deviations do not cancel out.* — A strong answer explains that signed deviations above and below the mean cancel to zero; absolute value keeps their nonnegative distances, including zero.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Both keepers have a mean of ___. Keeper A's MAD is ___, which means ___. Keeper B's MAD is ___, which means ___. The coach should pick ___ because ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.